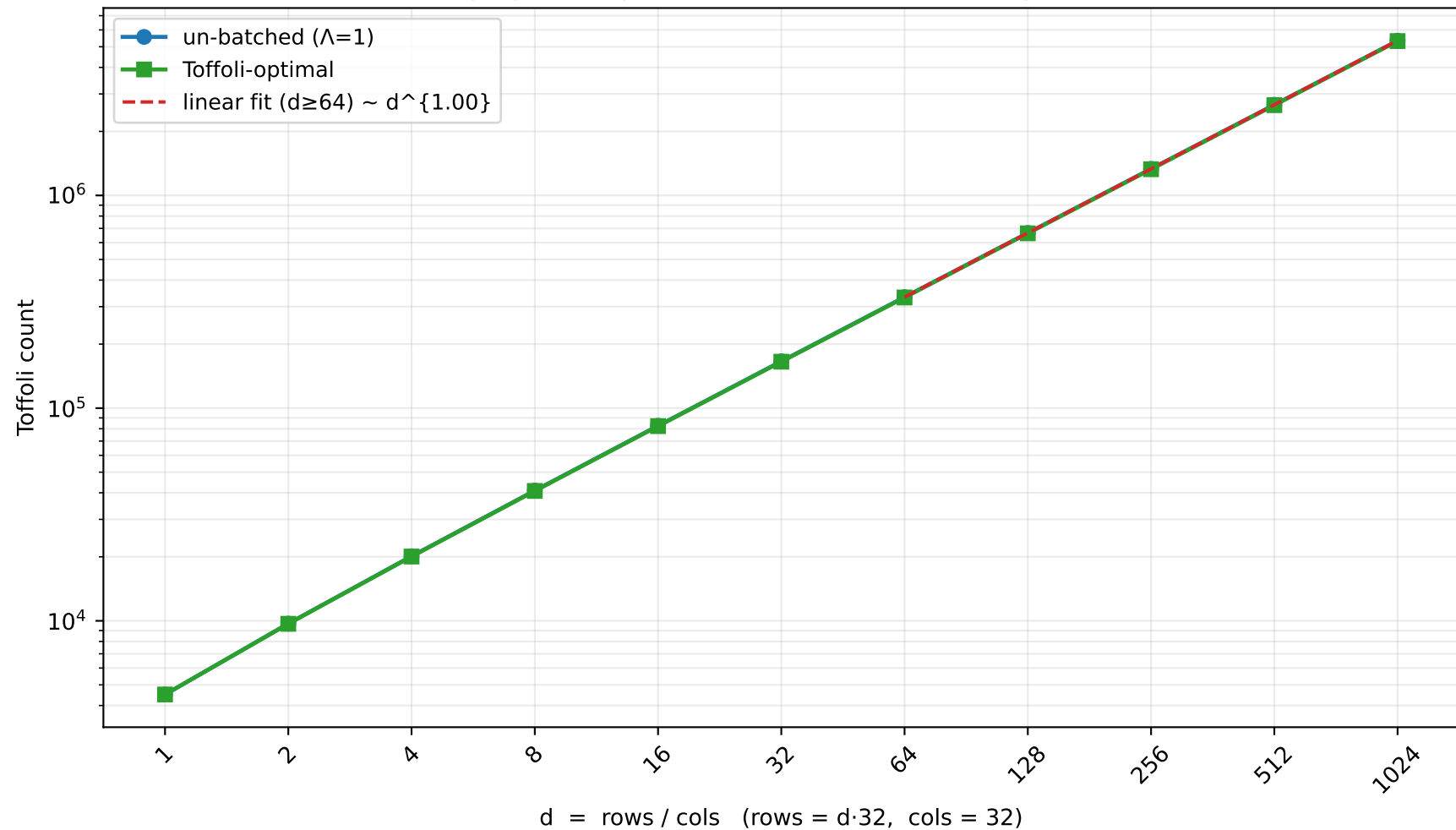
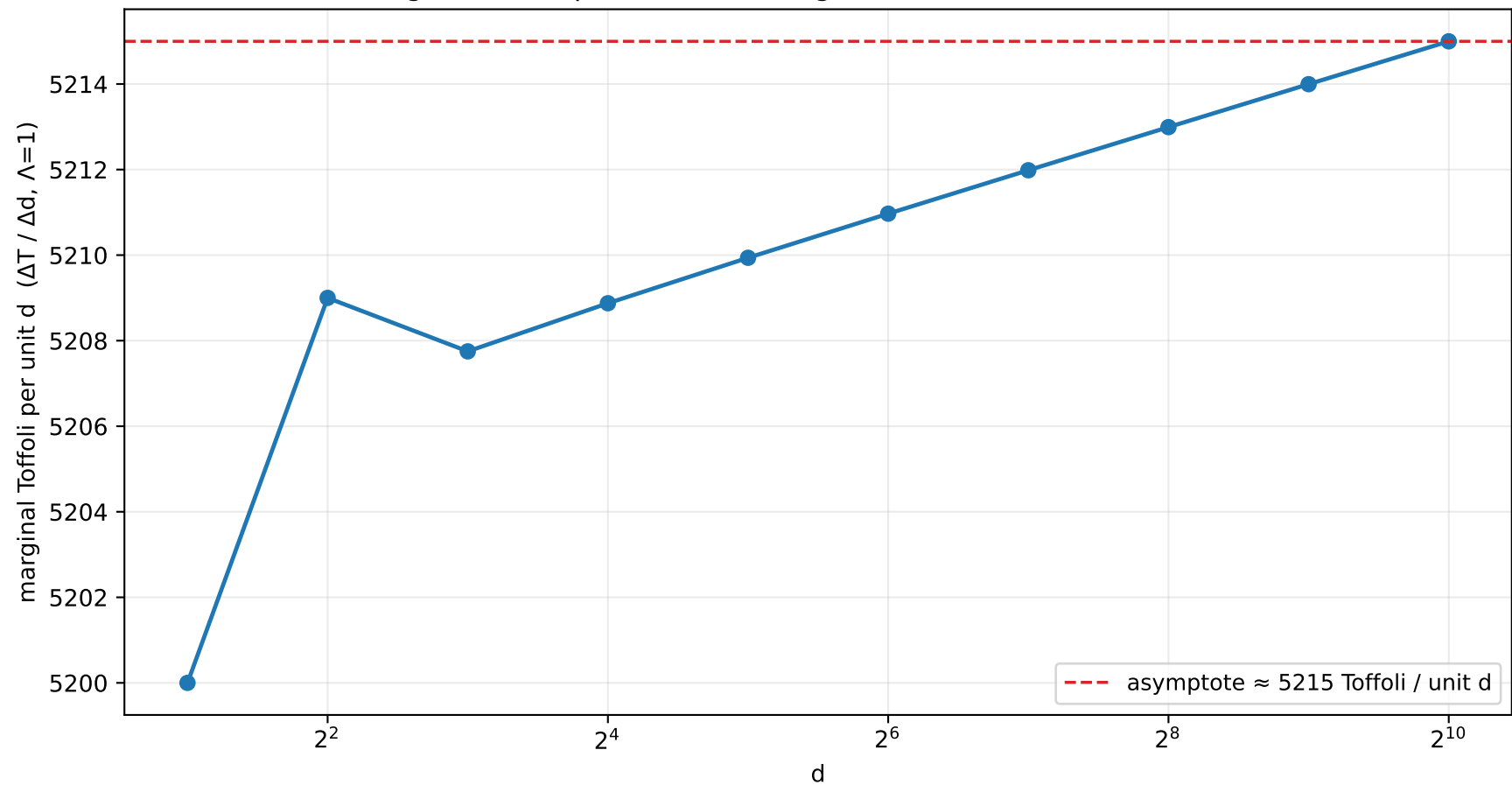


Interferometer isometry ($d \cdot N \times N$): Toffoli vs d ($N=32$, $b=32$)
asymptotically LINEAR in d (un-batched slope $\rightarrow 1$)



Marginal Toffoli per unit d converges to a constant $\Rightarrow T$ linear in d



Interferometer isometry Toffoli / qubits (N=32, b=32, single block)
 large-d fit ($d \geq 64$): $\Lambda=1$: $T \sim d^{1.001}$ opt: $T \sim d^{1.001}$

d	rows = d·N	cols (M)	T ($\Lambda=1$)	T (opt)	peak qubits
1	32	32	4,505	4,505	104
2	64	32	9,705	9,684	106
4	128	32	20,123	20,060	107
8	256	32	40,954	40,807	108
16	512	32	82,625	82,310	109
32	1024	32	165,984	165,333	110
64	2048	32	332,735	331,412	111
128	4096	32	666,302	663,635	112
256	8192	32	1,333,565	1,328,210	113
512	16384	32	2,668,348	2,657,617	114
1024	32768	32	5,338,427	5,316,944	115